**Fraud Risk Modelling (Credit Card)** **Executive Summary**

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**Objective:** My objective in this project is to build a production ready Fraud Risk solution using Machine Learning (ML) best practises. This solution is designed to assist financial institutions mitigate Fraud Risk through early detection of fraudulent activities.

### Exploratory Data Analysis (EDA)

The dataset used is European Credit Card Transactions dataset, it covers two days of transactions, provided by Kaggle. The dataset is severely imbalanced with fraud cases = **0.173%** of transactions i.e. 492 frauds cases vs. 284,315 non-frauds as shown in Figure 1 below. The total number of features is 30 features/columns, V1 to V28, Time and Amount.

I conducted an in-depth EDA highlighting the following key observations: The time feature has a multimodal distribution i.e. peaks of activity, several features including V13, V15, V18 and V19 have near-normal distribution. Some features have low variance/ concentrated while others show strong class separation. No missing values in the raw dataset. Overall, the EDA concluded that the problem is highly imbalanced and nonlinear, indicating the need for meticulous and targeted feature engineering. Tree-based ensemble models are most promising candidates based on the nature of the problem and the statistical behaviour of the features.

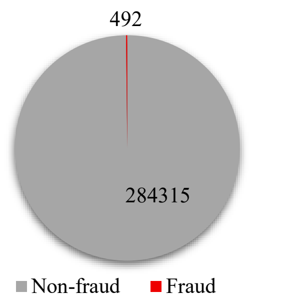


Figure 1: Class Imbalance

### Feature Engineering

I based my decisions and processes at this stage on EDA results. I conducted targeted (non-generic) feature engineering, by applying feature transformation according to each feature's statistical behaviour. For features with strong class separation, I applied non-monotonic transforms e.g. absolute and squaring to amplify the separation ability. I scaled features with low variance using MinMax scaling and converted Time feature into two features 'hour_of_day' and 'time_segment' to capture daily patterns.

The resulted engineered features empirically proved to be highly impactful as *40% of the final modes' most important predictors are engineered*, showing that targeted engineering improved patterns' detection.

### Modelling And Evaluation Workflow

Based on the EDA and the nature of the feature I selected and trained the following models: Logistic Regression as a baseline, Random Forest, CatBoost and LightGBM as advanced models. During the tuning stage, a hyperparameter search was conducted with 50 different combinations for each advanced model. After that, I conducted Out-of-Fold (OOF) Evaluation for all models. This OOF evaluation focuses on positive/fraud class performance metrics, including PR AUC. Based on OOF metrics the best performing model is the **tuned LightGBM** as it achieves the highest OOF PR AUC of 0.863, and a strong F1 score of 0.883 for fraud class.

### Calibration And Thresholding

Post OOF evaluation, I assessed LightGBM's probability reliability through calibration using Brier score, it achieved a noticeably low score of ~ **0.00037**. This indicates that the model produces realistic and reliable probabilities that can be interpreted as risk of fraud i.e. ***actual likelihood of fraud***. Subsequently, I selected validated and adjusted three probability thresholds to serve as Fraud Risk cut-offs for production API. The aim was to balance thresholds interpretability and coverage based on the OOF probabilities, which resulted in the following thresholds: *High Risk ≥ 0.80, Medium Risk 0.30 – 0.80 and Low Risk < 0.30*. Due to the well calibrated probabilities, Brier score ~ 0.00037, and the strong precision I have observed that the model produces few mid-range probability scores, effectively producing a near binary probability distribution, either fraud or non-fraud.

### Test Evaluation, Positive Class Focused

I evaluated all models on unseen test data and ranked them based on their test PR AUC score. I observed a small drop in test performance metrics compared to OOF metrics, this was expected due to generalisation in unseen data *e.g. LightGBM OOF PR AUC = 0.863 and test PR AUC = 0.857.*

Table 1 Final Production Model Test/Holdout data Metrics:

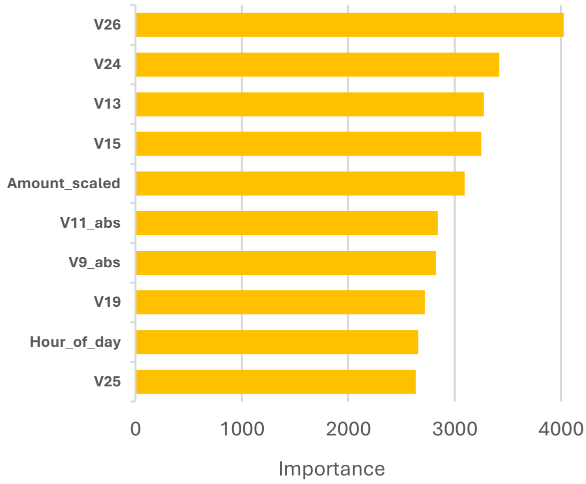
| Class | Precision | Recall | F1-score | Support | Balanced Accuracy | PR AUC |
|---|---|---|---|---|---|---|
| Fraud (Class 1) | 0.95 | 0.776 | 0.854 | 98 | 0.888 | 0.857 |
| Non fraud (Class 0) | 1 | 1 | 1 | 56864 | – | – |
| Macro Avg | 0.975 | 0.888 | 0.927 | 56962 | 0.888 | 0.857 |

The **Final positive class test metrics** for the production model are **PR AUC 85.7%, Precision = 95%, Recall = 77.6%**. I computed a normalized confusion matrix showing 77.6% true positive and 22.4% false negative for the positive/ fraud class. I further compared it with the confusion matrix of the 2nd best model to demonstrate the superiority of the selected model. For non-fraud class metrics, the matrix demonstrates a near-perfect true negative rate. The ranking of models based on PR AUC is consistent between OOF evaluation and Test evaluation, this confirms the appropriateness of production model selection.

**Feature Importance and Explainability**

In this section I computed production-model's feature importance based on gain, gain measures feature contribution in loss reduction during training. The list of top ten strongest predictors includes four engineered features, 40%, as shown in Figure 2, e.g. amount_scaled and hour_of_day. This demonstrate that the targeted engineering **did** increase class separability and interpretability.

Figure 2: Most Important Predictors




**Client facing API and Smoke Test**

I developed and tested a user API "fraud_risk_assessment()". I designed this API to produce business friendly outputs, in the form of table/DataFrame. This table presents fraud probability, risk level, risk indicator and an action code and text for each new transaction. While risk levels thresholds have been calibrated from a technical perspective and set to default values: *High ≥ 0.80, Medium 0.30 – 0.80 and Low < 0.30*, the API allows the client to change these thresholds based on business knowledge and risk appetite. After that I conducted a smoke test sampling test rows across all Risk tiers High/ Medium/ Low to assess the output comprehensively. An example of the output is shown in Table 2 below. This test verified schema and coverage, it also confirmed that the calculated risk levels matched the design, showcasing the API's effectiveness, interpretability and operational readiness.

Table 2 Example of Fraud Risk Assessment API output including Fraud probability, risk level, indicator and action.

| Transaction ID | Fraud Probability | Risk Level | Risk Indicator | Action code | Action Text |
|---|---|---|---|---|---|
| 26 | 1 | High | Red | BLOCK | Block transaction; initiate fraud protocol. |
| 93 | 0.42 | Medium | Amber | REVIEW | Flag for manual review. |
| 33 | 0.2 | Low | Green | ALLOW | No action; process normally. |

**Conclusion and Production Readiness**

Based on the EDA, the adequate training, the rigorous tuning and evaluation the recommended model is the tuned **LightGBM**. This model is wrapped in a user friendly three-tier API. Overall, this solution effectively handles the severe imbalance, fraud prevalence = 0.173%, provides calibrated realistic fraud risk probabilities and demonstrates a strong performance for the positive class: test PR AUC = 0.857, precision = 0.95 and recall = 0.776. I also addressed explainability via feature importance, further supporting material and results can be found in following directories:

- /results/plots/ for figures e.g. PR curves, importance, confusion matrices and etc).
- /results/tables/ for metric tables e.g., OOF/test metrics etc.